



MA.7.NSO.1.1

Benchmark

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

Benchmark Clarifications:

Clarification 1: Instruction focuses on building the Laws of Exponents from specific examples. Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.

Clarification 2: Problems in the form $\frac{a^n}{a^m} = a^p$ must result in a whole-number value for p .

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.1
- MA.7.GR.1.4
- MA.7.GR.2.2, MA.7.GR.2.3

Terms from the K-12 Glossary

- Base (of an exponent)
- Exponent (exponential form)
- Rational Number
- Whole Number

Vertical Alignment

Previous Benchmarks

- MA.6.NSO.3.3, MA.6.NSO.3.4

Next Benchmarks

- MA.8.NSO.1.3
- MA.8.AR.1.1
- MA.8.NSO.1.7

Purpose and Instructional Strategies

In grade 6, students evaluated positive rational numbers and integers with natural number exponents. In grade 7, students build on this to evaluate numerical expressions and generate equivalent numerical expressions with positive and negative rational numbers having whole-number exponents. In grade 8, students will reach procedural fluency with evaluating numerical expressions with positive and negative bases and integer exponents.

- Instruction includes allowing students to develop the Laws of Exponents based on patterns emerging from a series of examples related to each Law of Exponents (*MTR.5.1*).

- Develop understanding of the zero exponent by using multiplication to increase values and division to decrease values (*MTR.5.1*).
 - For example, to show decreasing values, $5^3 \div 5 = 5^2$ and then $5^2 \div 5 = 5^1$ (or 5) and then $5^1 \div 5 = 5^0$ (or $5 \div 5 = 1$).

Patterns in Exponents	
5^5	$5 \cdot 5 \cdot 5 \cdot 5 \cdot 5$
5^4	$5 \cdot 5 \cdot 5 \cdot 5$
5^3	$5 \cdot 5 \cdot 5$
5^2	$5 \cdot 5$
5^1	5
5^0	1

- Full expansion of exponents, limited to whole number exponents and rational number bases, should be modeled to help develop these patterns.

$$3^4 \cdot 3^3 = (3 \cdot 3 \cdot 3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^7$$

$$\frac{3^5}{3^3} = \frac{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot 3 \cdot 3}{\cancel{3} \cdot \cancel{3} \cdot \cancel{3}} = 3^2$$

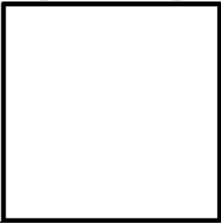
- Students should develop fluency with and without the use of a calculator when evaluating numerical expressions involving the Laws of Exponents.
- Instruction includes cases where students must work backwards as well as cases where the value of a variable must be determined (*MTR.3.1*). Students should use relational thinking as well as algebraic thinking.
 - For example, in $\frac{7^n}{7^2} = 343$, what is the value of n ? Students should ask themselves, “If I know that 343 is 7^3 , what value would n need to be so that $n - 2 = 3$?”
 - For example, in $(5^2)^n = 5^{10}$, what is the value of n ? Students should ask themselves, “What power would 5^2 be raised to equal 5^{10} ?”

Common Misconceptions or Errors

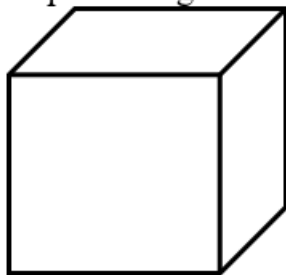
- Students may incorrectly conclude that squaring a number means to multiply by 2. Likewise, cubing may be mistaken as multiplying by 3. Use length to show doubling and area of a square to show an exponent of 2. Use of two-dimensional and three-dimensional manipulatives (*MTR.2.1*) may also help to emphasize squares and cubes versus increasing length.
- When finding the product or quotient of powers, students may incorrectly multiply or divide the bases, rather than only manipulating the exponents. Use full expansion of the exponential expression (*MTR.2.1*) to develop the laws.
- Students may incorrectly invert the product of powers and power of a power laws, wanting to multiply in the first and add in the latter.

Strategies to Support Tiered Instruction

- Instruction includes modeling the differences between doubling and squaring a value. Doubling a value would be represented by multiplying a given length by 2 whereas squaring a number would be represented by the area of a square with a given length.
 - For example, students can be given the table below to show how the left column doubles the length whereas the right column squares a length.

Doubling	Squaring
Given length of 5 _____	Given length of 5 _____
Representing $5 \cdot 2$ _____	Representing 5^2 

- Instruction includes modeling the differences between tripling or cubing a value. Tripling a value would be represented by multiplying a given length by 3, whereas cubing a number would be represented by the volume of a cube with a given length.
 - For example, students can be given the table below to show how the left column triples a length whereas the right column cubes a length.

Doubling	Cubing
Given length of 5 _____	Given length of 5 _____
Representing $5 \cdot 3$ _____ _____ _____	Representing 5^3 

- The teacher provides opportunities for students who incorrectly apply the Laws of Exponents when generating equivalent numerical expressions to use full expansion of exponential expression as an additional step to visually represent the emerging patterns.
 - For example, the expression $4^3 \cdot 4^2$ can be expanded to $(4 \cdot 4 \cdot 4) \cdot (4 \cdot 4)$ which is equivalent to $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$ or 4^5 . This helps illustrate the product of powers Law.

- Teacher creates and posts an anchor chart with visual representations of the base of an exponent, exponent and any factor(s), then encourages students to utilize the anchor chart to assist in correct academic vocabulary when evaluating exponential expressions.
 - For example, the teacher can highlight different parts of the expression $4^3 \cdot 4^2$ in different colors to visualize the **base**, **exponents** and **factors**: $4^3 \cdot 4^2$.
- Instruction includes co-creating a graphic organizer for each of the Laws of Exponents to include the name of the law, an original exponential expression, an equivalent expansion of the expression, the equivalent simplified expression, and a generalized rule for each of the laws.
 - For example, a graphic organizer with some of the Laws is shown.

Law	Original Exponential Expression	Possible Equivalent Expansion(s) of Expression	Equivalent Simplified Expression	Generalized Rule
Power of a Power	$(4^3)^2$	$(4 \cdot 4 \cdot 4)^2$ $(4 \cdot 4 \cdot 4)(4 \cdot 4 \cdot 4)$ $(4^3)(4^3)$	4^6	$(a^m)^n = a^{m \cdot n}$
Power of a Quotient	$\left(\frac{4}{5}\right)^4$	$\frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5} \cdot \frac{4}{5}$ $\frac{4 \cdot 4 \cdot 4 \cdot 4}{5 \cdot 5 \cdot 5 \cdot 5}$	$\frac{4^4}{5^4}$	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$

Instructional Tasks

Instructional Task 1 (MTR.5.1)

Complete the table by using numeric examples to develop a rule for each of the situations given. Then confirm or adjust your hypothesis as one of the Laws of Exponents.

	Numeric Examples	Rule Hypothesis	Confirmation or Revision	Law of Exponents
$a^m \cdot a^n$	$3^2 \cdot 3^4 = 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 3^6$ $2^3 \cdot 2^4 =$			
$\frac{a^m}{a^n}$	$\frac{4^5}{4^8} = \frac{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4}{4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4} = \frac{4 \cdot 4}{1} = 4^2$ $\frac{6^6}{6^6} = \frac{6 \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6}{6 \cdot 6 \cdot 6 \cdot 6 \cdot 6 \cdot 6} =$			
$(a^m)^n$	$(5^4)^5 =$			
$(ab)^m$				
$\left(\frac{a}{b}\right)^m$				

Instructional Items

Instructional Item 1

Generate an equivalent expression.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

Instructional Item 2

Evaluate the following expression.

$$\left(\left(\frac{1}{4}\right)^2\right)^3 \cdot \left(\frac{1}{4}\right)^3$$

Instructional Item 3

Evaluate the following expression.

$$\frac{(8^4)^3 \cdot 5^2 \cdot 5^3}{8^7 \cdot (8 \cdot 5)^4}$$

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*